

1 Evaluation Methodology

1.1 Step 1: LLM Stability Verification

Before using LLM judgments as ground truth, we verify that the LLM produces *consistent* results across repeated calls, regardless of correctness. We run each issue through the LLM five times independently and measure two complementary stability metrics.

Set agreement S_{set} measures the overlap of positive responses across runs:

$$S_{\text{set}} = \frac{|\bigcap_{i=1}^m A_i|}{|\bigcup_{i=1}^m A_i|} \quad A_i = \{\text{issues labelled Yes in run } i\} \quad (1)$$

where $m = 5$ is the number of runs. $S_{\text{set}} = 1$ means every run labels exactly the same set of issues as *Yes*; lower values indicate increasing disagreement.

Normalised entropy S_{entropy} measures per-issue consistency:

$$S_{\text{entropy}} = 1 - \bar{H} \quad \bar{H} = \frac{1}{n} \sum_{j=1}^n H_j \quad (2)$$

where for each issue j , $H_j = -p_j \log_2 p_j - (1 - p_j) \log_2 (1 - p_j)$ is the binary entropy of its m responses (p_j = fraction of *Yes* answers), and n is the total number of issues. $S_{\text{entropy}} = 1$ means all issues receive identical answers across all runs.

Table 1 compares the two methods on the 200 retained issues, using five runs per method.

Table 1: LLM Stability Metrics for AgentBug-Smith (Ours) vs. Raw AI API ($m = 5$ runs)

Method	S_{set}	S_{entropy}
AgentBug-Smith (Ours)	0.9158	0.9378
Raw AI API	0.7208	0.8929

The **AgentBug-Smith (Ours)** method achieves $S_{\text{set}} = 0.9158$ and $S_{\text{entropy}} = 0.9378$, substantially outperforming the **Raw AI API** baseline ($S_{\text{set}} = 0.7208$, $S_{\text{entropy}} = 0.8929$). This confirms that the agent-issue taxonomy and structured prompt used by AgentBug-Smith produce markedly more consistent judgments across repeated LLM calls than a direct, unstructured prompt.

1.2 Step 2: Sampling Strategy

From the full dataset of $N = 200$ labelled issues, we select $n = 132$ for detailed human annotation. The sample size is determined by Cochran’s formula for proportions:

$$n_0 = \frac{z^2 p(1-p)}{e^2} \quad n = \frac{n_0}{1 + \frac{n_0 - 1}{N}} \quad (3)$$

where $z = 1.96$ (95% confidence), $p = 0.5$ (maximum variance assumption), $e = 0.05$ (target margin of error), and $N = 200$ (population size). This yields $n_0 \approx 384.2$, which after finite-population correction gives $n \approx 131.8$, rounded up to $n = 132$.

1.3 Step 3: Human Annotation and Ground-Truth Construction

The 132 sampled issues are independently labelled by two annotators (the first author, **Yiming**, and a domain expert, **Alfin**) using the agent-issue taxonomy in [1]. Each annotator classifies every issue as *Yes* (agent issue) or *No* (not an agent issue).

Inter-annotator agreement. We measure agreement with Cohen’s κ :

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (4)$$

where

$$p_o = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[a_i = b_i] \quad p_e = \sum_{c \in \{Y, N\}} p_c^{(a)} p_c^{(b)} \quad (5)$$

p_o is observed agreement, p_e is expected agreement by chance, a_i and b_i are the labels assigned by annotators a (Yiming) and b (Alfin) to issue i , and $p_c^{(a)}$, $p_c^{(b)}$ are the marginal proportions of class c for each annotator.

Table 2: Human Annotation Statistics ($n = 132$)

Metric	Value
Total labelled issues	132
Yiming labels — Yes / No	129 / 3
Alfin labels — Yes / No	129 / 3
Human Golden — Yes / No	128 / 4
Observed agreement p_o	128/132 = 0.9697
Chance agreement p_e	$0.9654 \times 0.9654 + 0.0346 \times 0.0346 = \mathbf{0.9556}$
Cohen’s κ	0.3178
Yiming accuracy vs. Human Golden	131/132 = 0.9924
Alfin accuracy vs. Human Golden	129/132 = 0.9773

The annotator contingency table is

	Alfin: Yes	Alfin: No
Yiming: Yes	127	2
Yiming: No	2	1

The off-diagonal counts are balanced (2 vs. 2), indicating no directional tendency for one annotator to assign more positive labels than the other. Positive agreement is high,

$$P_{\text{pos}} = \frac{2(127)}{2(127) + 2 + 2} = 0.9845,$$

whereas negative agreement is

$$P_{\text{neg}} = \frac{2(1)}{2(1) + 2 + 2} = 0.3333.$$

Thus, the annotators agree strongly on positive cases, while the small number of negative cases is intrinsically less stable.

Adjudication. After independent annotation, the disagreements are reviewed and Alfin assigns the final adjudicated label, denoted **Human Golden**. This label is used as the reference for downstream validation. Because Alfin performs the final adjudication, the reported agreement

between Alfin’s initial labels and Human Golden is descriptive rather than an independent accuracy estimate.

Interpretation standard for κ . We use the commonly cited Landis–Koch interpretation ranges [2]:

Table 3: Landis–Koch Interpretation of Cohen’s κ

Cohen’s κ range	Strength of agreement
$\kappa < 0.00$	Poor
0.00–0.20	Slight
0.21–0.40	Fair
0.41–0.60	Moderate
0.61–0.80	Substantial
0.81–1.00	Almost perfect

Our result, $\kappa = 0.318$, falls within the *fair agreement* range (0.21–0.40). However, the raw agreement $p_o = 96.97\%$ is very high. This apparent paradox is a known artefact of Cohen’s κ under severe class imbalance: when one class dominates (129 of 132 initial labels are *Yes* for each annotator), even a small number of disagreements sharply reduces κ [3]. Therefore, the categorical range should be treated as descriptive rather than as an absolute assessment of annotation quality. As a prevalence-robust sensitivity check, Gwet’s AC1 is 0.9683, which is consistent with the observed agreement and confirms that the modest Cohen’s κ is primarily caused by prevalence rather than broad annotator disagreement.

1.4 Step 4: Ours vs. Raw LLM against Human Golden

Human Golden on the $n = 132$ sample comprises 128 *Yes* and 4 *No*. We compare two predictors against this reference.

- **Ours** (AgentBug-Smith): the taxonomy-guided filter; every sampled issue already received *Yes*.
- **Raw LLM**: gpt-4.1-mini, five independent votes on issue title and body only (no patch, no ai_judgment, no taxonomy), aggregated by majority.

Confusion counts.

TP	=	#(pred = Yes, Golden = Yes)	Acc	=	(TP + TN)/ n
FP	=	#(pred = Yes, Golden = No)	P	=	TP/(TP + FP)
TN	=	#(pred = No, Golden = No)	R	=	TP/(TP + FN)
FN	=	#(pred = No, Golden = Yes)	Spec	=	TN/(TN + FP)
			F_1	=	2PR/(P + R)

Because Golden has only four negatives, $FP + TN = 4$ and $TP + FN = 128$ for both methods.

Table 4: Ours vs. Raw LLM: confusion *counts* against Human Golden ($n = 132$)

Method	TP	FP	TN	FN
Ours	128	4	0	0
Raw LLM	70	4	0	58

Both methods label all four Golden-*No* issues as *Yes* ($FP = 4$, $TN = 0$), so specificity is 0 in either case. The gap is recall: Ours never misses a Golden-*Yes* issue ($FN = 0$, $R = 100\%$),

Table 5: Ours vs. Raw LLM: cell percentages of $n = 132$ (TP% = TP/ n , etc.)

Method	TP%	FP%	TN%	FN%
Ours	128/132 = 96.97%	4/132 = 3.03%	0%	0%
Raw LLM	70/132 = 53.03%	4/132 = 3.03%	0%	58/132 = 43.94%

Table 6: Ours vs. Raw LLM: confusion *rates* against Human Golden

Method	Acc	Precision	Recall	Spec	F_1
Ours	128/132 = 96.97%	128/132 = 96.97%	128/128 = 100%	0/4 = 0%	98.46%
Raw LLM	70/132 = 53.03%	70/74 = 94.59%	70/128 = 54.69%	0/4 = 0%	69.31%

whereas Raw LLM misses $58/128 = 54.69\%$ of them. Accuracy therefore falls from 96.97% (Ours) to 53.03% (Raw LLM). Ours’ recall of 100% holds only on this already-filtered sample, where the method predicts *Yes* for every case.

References

- [1] Eamin et al., *AgentBug-Smith: A Taxonomy of Agent Issues in LLM-based Agent Systems*, 2025.
- [2] Landis, J. R., and Koch, G. G., “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [3] Feinstein, A. R., and Cicchetti, D. V., “High agreement but low kappa: II. Resolving the paradoxes,” *Journal of Clinical Epidemiology*, vol. 43, no. 6, pp. 551–558, 1990.